

Bidirectional LSTM Sequence Baseline

Input Sequence Tensor

Shape: $[B, T = 32, D = 117]$



LayerNorm(117)

Feature standardization across joints



2-Layer Bidirectional LSTM

hidden_dim = 96 per direction ($\vec{h}_t \in \mathbb{R}^{96}, \overleftarrow{h}_t \in \mathbb{R}^{96}$)

Forward + Backward = 192 | Dropout = 0.3

Output: $[B, T = 32, 192]$



Temporal Global Average Pooling (GAP)

pooled = $\frac{1}{T} \sum_{t=1}^T [\vec{h}_t \parallel \overleftarrow{h}_t] \rightarrow [B, 192]$



Two-Stage MLP Classifier Head

Dropout(0.3) $\rightarrow$ Linear(192 $\rightarrow$ 64) $\rightarrow$ ReLU
 $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(64 $\rightarrow$ 22) $\rightarrow$ Softmax
Output: 22 Exercise Class Probabilities $P \in \Delta^{21}$